

assign6n

October 7, 2026

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[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# =====
# Q1 - Mathematical Functions
# =====

M = float(input("Enter the value of M: "))

x = np.linspace(-10, 10, 100)

y1 = M * x**2
y2 = M * np.sin(x)

plt.figure(figsize=(8, 5))
plt.plot(x, y1, label="M*x^2")
plt.plot(x, y2, label="M*sin(x)", linestyle="--")
plt.xlabel("x")
plt.ylabel("y")
plt.title("Mathematical Functions")
plt.legend()
plt.grid(True)
plt.show()

# =====
# Q2 - Subjects and Scores
# =====

data = {
    "Subject": ["Maths", "Python", "DBMS", "DAA", "Statistics"],
    "Score": [85, 92, 78, 88, 95]
}

df = pd.DataFrame(data)
```

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print(df)

plt.figure(figsize=(8, 5))

ax = sns.barplot(
    x="Subject",
    y="Score",
    data=df,
    palette="viridis"
)

for container in ax.containers:
    ax.bar_label(container)

plt.title("Subject-wise Scores")
plt.xlabel("Subjects")
plt.ylabel("Scores")
plt.grid(axis="y")
plt.show()

# =====
# Q3 - Random Dataset and 2x2 Subplots
# =====

np.random.seed(1024170366)

data = np.random.randn(50)
x = np.arange(1, 51)

cumulative_sum = np.cumsum(data)
noise = np.random.randn(50)

fig, axes = plt.subplots(2, 2, figsize=(12, 8))

# Cumulative sum
axes[0, 0].plot(x, cumulative_sum)
axes[0, 0].set_title("Cumulative Sum")
axes[0, 0].set_xlabel("Index")
axes[0, 0].set_ylabel("Cumulative Sum")
axes[0, 0].grid(True)

# Scatter plot with noise
axes[0, 1].scatter(x, data + noise)
axes[0, 1].set_title("Scatter Plot with Random Noise")
axes[0, 1].set_xlabel("Index")

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axes[0, 1].set_ylabel("Value")
axes[0, 1].grid(True)

# Random data
axes[1, 0].plot(x, data)
axes[1, 0].set_title("Random Data")
axes[1, 0].set_xlabel("Index")
axes[1, 0].set_ylabel("Value")
axes[1, 0].grid(True)

# Histogram
axes[1, 1].hist(data)
axes[1, 1].set_title("Distribution of Random Data")
axes[1, 1].set_xlabel("Value")
axes[1, 1].set_ylabel("Frequency")
axes[1, 1].grid(True)

plt.tight_layout()
plt.show()

# =====
# Q4 - Company Sales Data
# =====

url = "https://raw.githubusercontent.com/AnjulaMehto/MCA/main/
↳company_sales_data.csv"

sales = pd.read_csv(url)

print(sales.head())

# Q4(1) - Total Profit of All Months

plt.figure(figsize=(10, 5))

sns.lineplot(
    x="month_number",
    y="total_profit",
    data=sales,
    marker="o"
)

plt.title("Total Profit of All Months")
plt.xlabel("Month Number")
plt.ylabel("Total Profit")

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plt.grid(True)
plt.show()

# Q4(2) - All Product Sales

products = [
    "facecream",
    "facewash",
    "toothpaste",
    "bathingssoap",
    "shampoo",
    "moisturizer"
]

plt.figure(figsize=(12, 6))

for product in products:
    sns.lineplot(
        x="month_number",
        y=product,
        data=sales,
        marker="o",
        label=product
    )

plt.title("Monthly Product Sales")
plt.xlabel("Month Number")
plt.ylabel("Sales")
plt.legend()
plt.grid(True)
plt.show()

# Q4(3) - Bar Chart for All Features

plt.figure(figsize=(14, 7))

sales_melted = sales.melt(
    id_vars="month_number",
    var_name="Feature",
    value_name="Value"
)

sns.barplot(
    x="Feature",
    y="Value",

```

```

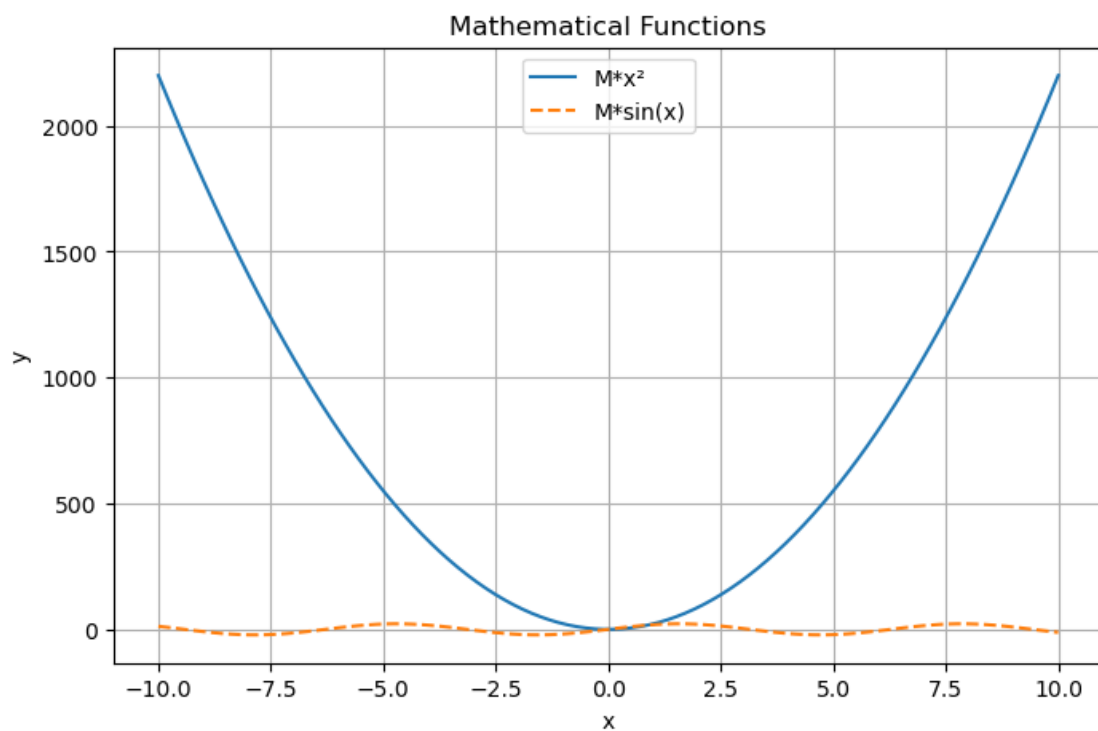
    data=sales_melted
)

plt.title("Bar Chart for All Features")
plt.xlabel("Features")
plt.ylabel("Values")
plt.xticks(rotation=45)
plt.grid(axis="y")
plt.show()

```

Matplotlib is building the font cache; this may take a moment.

Enter the value of M: 22



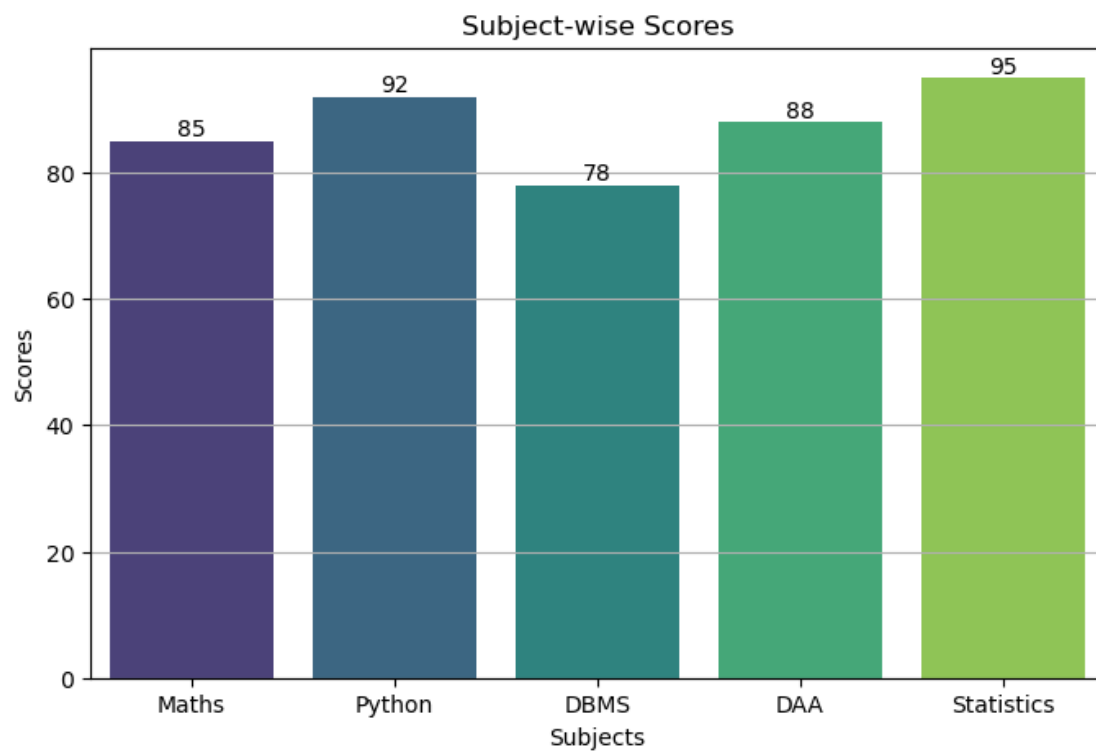
	Subject	Score
0	Maths	85
1	Python	92
2	DBMS	78
3	DAA	88
4	Statistics	95

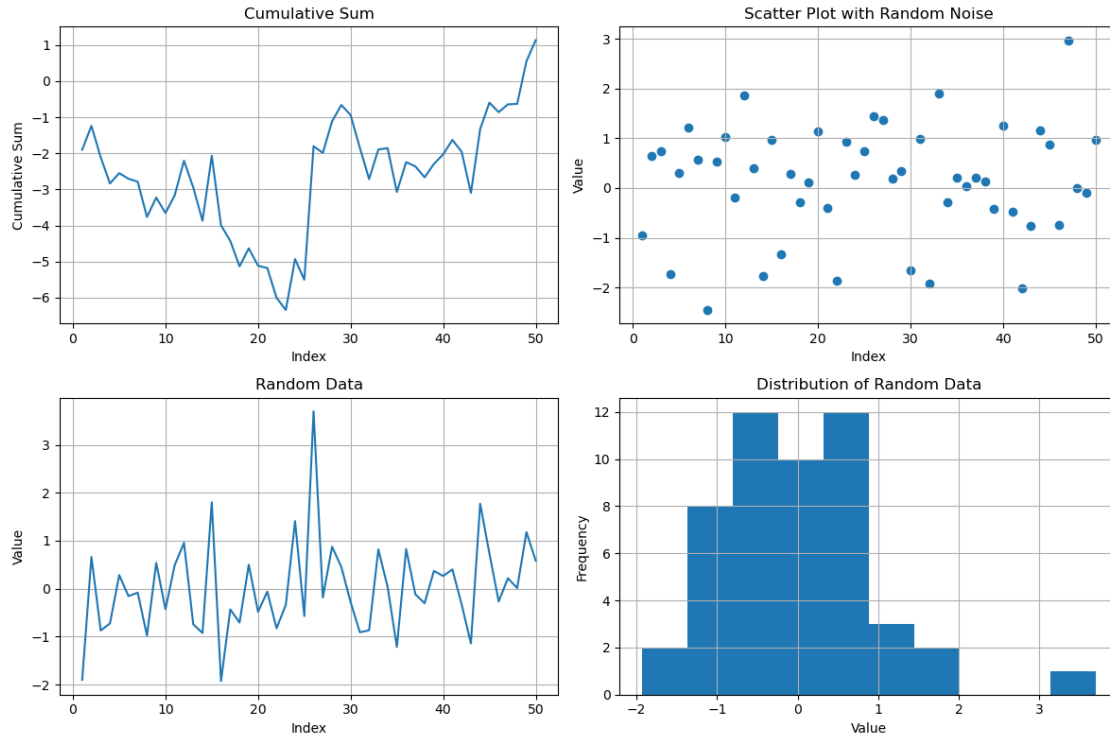
/tmp/ipykernel_249/2552521389.py:43: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same

effect.

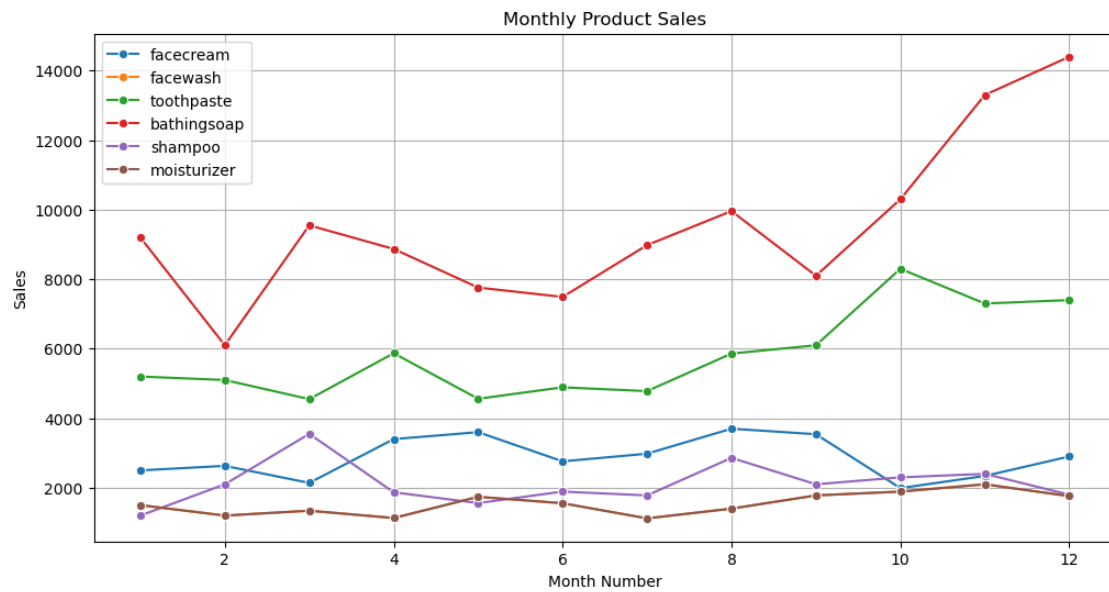
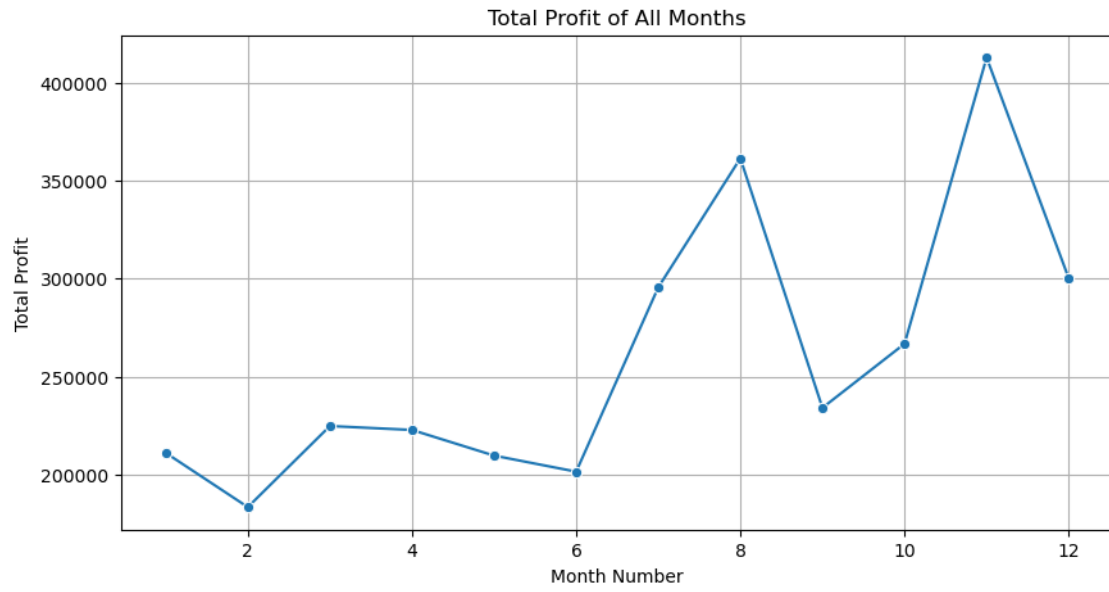
```
ax = sns.barplot(
```

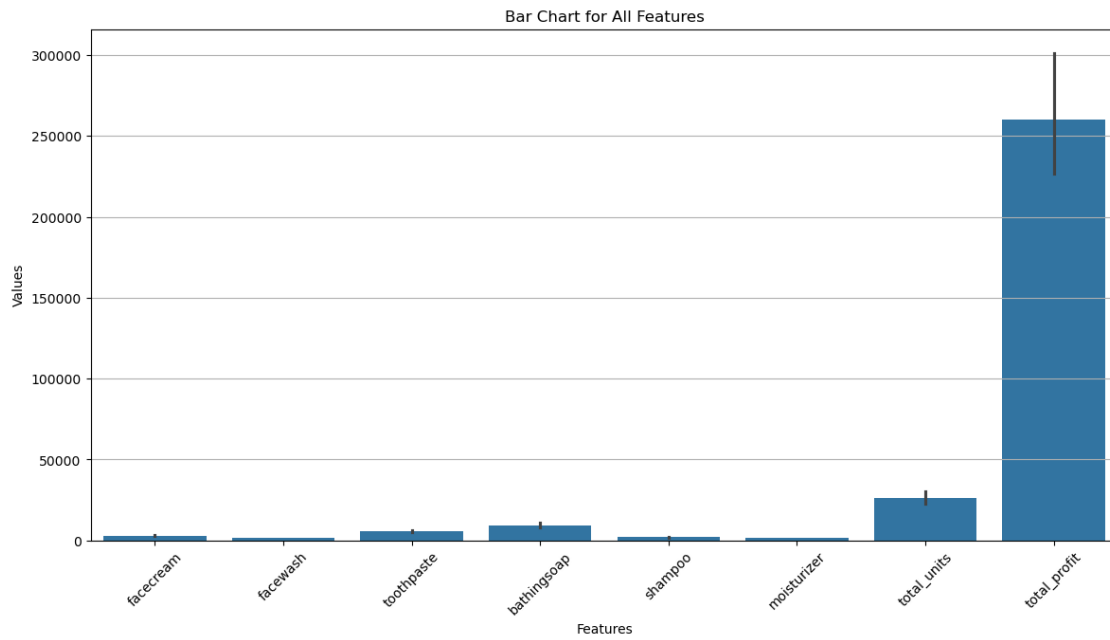




	month_number	facecream	facewash	toothpaste	bathingsoap	shampoo	\
0	1	2500	1500	5200	9200	1200	
1	2	2630	1200	5100	6100	2100	
2	3	2140	1340	4550	9550	3550	
3	4	3400	1130	5870	8870	1870	
4	5	3600	1740	4560	7760	1560	

	moisturizer	total_units	total_profit
0	1500	21100	211000
1	1200	18330	183300
2	1340	22470	224700
3	1130	22270	222700
4	1740	20960	209600





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